Cognitive Profiling Prediction

Online Hackathon

Objective

The goal of this task is to build a model that analyzes video clips and predicts the set of actions occurring at specific timestamps. Each timestamp in a video may be associated with **one or more action labels**, making this a **multi-label classification** problem.

The possible action labels include:

- **Advancing**
- **Retreating**
- **Enclosing**
- **Spreading**
- **Rising**
- **Descending**
- **Directing**
- **Indirecting**
- **Increasing Pressure**
- **Decreasing Pressure**
- **Accelerating**
- **Decelerating**

Participants are expected to model both spatial and temporal dynamics to accurately detect and classify overlapping or co-occurring actions in time.

Dataset

- **train.csv** - Contains metadata and multi-label annotations for the training videos. Each row specifies the video, timestamp, and associated actions.

- **train/** - A directory containing **11 video files** used for training. Some supplementary labeled image data (e.g., from celebrity sources) may also be included in this folder to aid learning.

- **test_submission.csv** - A sample submission file listing the required format for predictions on the test set. Each row represents a video and timestamp where the model must predict the relevant actions.

- **test/** - A directory containing **2 test video files.**

Video Duration: Each video is approximately **~10 to ~40 minutes long**

Dataset Example

train.csv (15.37 kB)

Detail | Compact | Column — 10 of 14 columns

About this file

This file does not have a description yet.

Filename	Time	Advancing	Retreating	Enclosing	Spreading
s03_smiley 28% s01_smiley 23% Other (181) 49%	332 unique values	0 – 1	0 – 1	0	
s01_smiley	t0405	0	0	0	
s01_smiley	t0355	0	1	1	
s01_smiley	t0448	0	0	1	
s01_smiley	t0512	0	0	1	
s01_smiley	t0548	0	0	0	
s01_smiley	t0637	0	0	0	
s01_smiley	t0710	0	0	0	
s01_smiley	t0717	0	1	1	
s01_smiley	t0743	0	0	0	
s01_smiley	t0747	0	0	0	

Dataset

train.csv (15.37 kB)

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s01_smiley	t0512	0	0		
s01_smiley	t0548	0	0		
s01_smiley	t0637	0	0	0	1
s01_smiley	t0710	0	0	0	1
s01_smiley	t0717	0	1	1	0
s01_smiley	t0743	0	0	0	1
s01_smiley	t0747	0	0	0	1

Timestamp Format

The **Time** column (or timestamp feature) uses a string format such as t0405, which represents the time **4 minutes and 5 seconds** into the video. The format follows this pattern:

- tMMSS
  MM = Minutes (zero-padded to 2 digits)
  - SS = Seconds (zero-padded to 2 digits)

Example:

- t0930 → 9 minutes and 30 seconds
- t0015 → 0 minutes and 15 seconds
- t1035 → 10 minutes and 35 seconds

Evaluation

Evaluation Metric: **sample-average F1**

Public Leaderboard: 50%

Private Leaderboard: 50%

Rules

- การแข่งขันเป็นรูปแบบ**บ้าน** (บ้านหรือศูนย์มหาวิทยาลัย)
- โดยผู้เข้าแข่งขันจะต้องตั้งชื่อทีมให้ถูกต้องตามที่กำหนด คือ "ชื่อบ้าน" เช่น "จามจุรี"
- ห้ามใช้ Third-party API
- ห้ามใช้ Private Dataset

Important Information

ระยะเวลาการแข่ง : Mon 16 June 12:00 - Fri 20 June 08:00

จำนวนครั้งในการส่ง : 6 ครั้ง/วัน, ตัดรอบ 07.00 น.

Kaggle link

https://www.kaggle.com/t/62d4a3dfb52c4264a3f3a190eb44621d

ประกาศรายชื่อทีม Pitching Fri 20 June เวลา 08.30 น.

Q & A